Module 3

Discrete-Time Dynamic Programming and Reinforcement Learning

In this document we present several MATLAB and Simulink engineering design projects relevant to dynamic programming and reinforcement learning in the discrete-time domain according to the material presented in Modules 1, 2, and 3.

MATLAB and Simulink Reinforcement Learning Design Projects

In this section, we present several MATLAB discrete-time RL design projects, which should be implemented as closely as possible according to the MATLAB reinforcement learning implementation structure presented in Figure 1.1 in Module 1 using MATLAB live scripts and live functions, eventually using the Simulink toolbox.

Design Project 3.1. RL Algorithm for the Electric Islanded Microgrid System

In this design project, students use MATLAB and its live scripts and live functions (and if needed Simulink) to demonstrate the RL solution (see block diagram in Figure 1.1 in Module 1) to the LQ optimal control problem in the discrete-time domain using the Hewer algorithm presented in Section 3.5 in Module 3.

Project Report Format: The project report is supposed to be typed, and it must contain all standard technical paper main parts: introduction (about the general methodology used), problem formulation (what you plan to do and what you are supposed to achieve) main results obtained (obtained simulation results, tables, and figures), discussion/comments (on obtained results and some observations), conclusions, appendix (with MATLAB or any other software code), references. The report should contain a brief description of the Hewer policy iteration algorithm with analytical presentation of all important formulas, equations, required conditions (assumptions).

Project Data: The project considers an islanded electric microgrid system whose continuous-time space matrices are presented in Problem 2.9 of Module 2. Discretize this system with the

sampling rate $T_s = 0.01$. Assume that the penalty matrices in the corresponding LQ optimal control problem are identities, that is, given by $R_1 = I_4, R_2 = I_2$.

Project Task: Running policy iterations of the Hewer algorithm generate a table that shows how the approximate performance values converge per iteration to the optimal values. In general, you need to run only five to six iterations, in rare case nine or ten iterations due to the quadratic convergence of the algorithm. Plot on one figure all approximates feedback controls obtained for $k = 1, 2, 3, 4, 5$, and on another figure plot the approximate system state trajectories for a selected (the one that performs the worst) state variable for $k = 1, 2, 3, 4, 5$.

Hint: First check a simple MATLAB demo code done for an inverted pendulum presented in this project assignment. Use the MATLAB Live scripts with two sliders for the number of policy iterations and for the length of time interval of optimization.

Design Project 3.2. RL Algorithm for the Wind Turbine System

A continuous-time linear wind turbine mathematical model state space matrices are presented in Problem 2.10 in Module 2. Discretize this model with the sampling rate $T_s = 0.01$. To the obtained discrete-time linear system apply the Hewer reinforcement learning algorithm following the project formulation and data presented in Problem 2.10 of Module 2 and Design Project 3.1 from Module 3. Write a live script code with corresponding live functions, including interactive involvement of Simulink whose role is to simulate the wind turbine dynamic response.

Design Project 3.3. RL Algorithm for a Satellite Optimal Control

Consider the satellite optimal control problem whose continuous-time state space matrices are given in Problem 2.10 of Module 2. Discretize this model with the sampling rate $T_s = 0.05$. Repeat all steps outlined of Design Project 3.1 from Module 3 for reinforcement learning optimal control of the satellite mathematical model. Write a live script code with corresponding live functions, including their interactive cooperation with Simulink whose role is to simulate the dynamic response of considered satellite model.

Design Project 3.4. RL Algorithm for an F-15 Aircraft Optimal Control

Consider the satellite optimal control problem whose continuous-time state space matrices are given in Problem 2.12 in Module 2. Discretize this model with the sampling rate $T_s = 0.05$. Repeat all steps outlined of Design Project 3.1 from Module 3 for RL optimal control of this aircraft. Implement the reinforcement learning mechanism according to Figure 1.1 from Module 1 using Simulink.

Design Project 3.5. RL Algorithm for a Drone Yaw Angle Optimal Control

Consider a drone yaw angle control continuous-time transfer function considered in Exercise 2.1 of Module 2. Obtain its state space form. Discretize the state space model obtained with the

sampling rate $T_s = 0.01$. Repeat all steps outlined of Design Project 3.1 from Module 3 for RL optimal control drone yaw angle control. Implement the RL mechanism according to Figure 3.1 from Module 3 using Simulink.